

graphical programming, it is the data flow between block diagram nodes that controls the execution order. The visual nature of LabVIEW makes it straightforward for programming novices to create applications, especially when using features such as express VIs, which replace the coding tasks with a dialogue box configuration.

Front panel—the graphical user interface

LabVIEW provides the ability to build graphical user interfaces, which it calls front panels to provide an analogy with physical instruments. As the main focus of LabVIEW is science and engineering, in addition to standard user interface objects such as 'OK' buttons and text entry boxes, there are lots of items specifically aimed at these areas.

There is a low-cost LabVIEW Student Edition aimed at educational institutions for learning purposes (which is included in the DVD). There is also an active community of LabVIEW users who communicate through several e-mail groups and Internet forums, thus making sure that users can fiddle around with the system and still not get completely lost.

Summary

The distinguishing feature when compared to other programming languages is that LabVIEW uses graphics instead of text listings for its source code. LabVIEW is very popular with scientists and engineers, who do not necessarily have a background in software programming, due to this graphical coding and straightforward integration with a wide range of measurement instruments and control equipment. With the updates for the software pouring in, LabVIEW is all set to be the industry's darling among system design software for the time being.

How to install LabView: Student's edition

There is a proper order for installing National Instruments' software based on how the individual pieces work together. Here is the order in which to

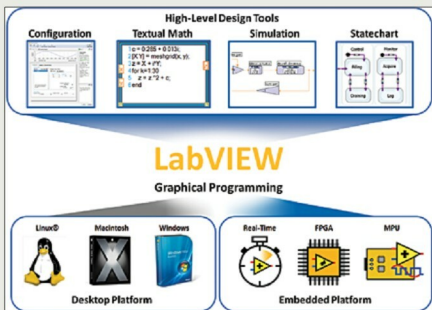


Fig. 3: NI LabVIEW

install the suite:

1. Development system (LabVIEW)
2. Software toolkits and modules (LabVIEW Control Design and Simulation Toolkit)
3. Hardware drivers (NI-ELVISmx Device Driver)

The development system is installed first because it is the base or platform onto which everything else is installed. The modules and toolkits are then installed because they are development system dependent, and are added in place of the development system like the functions palette of LabVIEW. Last of all, the drivers are installed along with the driver support for the development system and toolkits.

If they are not installed in this order, the dependent features will not realise that the development system is present, and thus not appear in the development system environment.

Once the DVD is inserted into the system, the program will automatically start the installation process. Note, if 'autoplay' is disabled, proceed to the DVD drive, double click on the 'setup.exe' file and follow the instructions to start the suite install process.

After installing LabVIEW, Student Edition, it is the right time to install hardware drivers. You can obtain the

hardware drivers to use with LabVIEW from the DVD accompanying the magazine. After correctly installing the drivers and restarting the system, you are now ready to launch LabVIEW. Have fun with the software!

Getting started with LabVIEW

Launch LabVIEW by click-starting on your Windows toolbar and navigating to 'All Programs.' From the program directory, click 'National Instruments LabVIEW 2011 SP1.' The 'Getting Started' window is divided in half from left to right. On the left is the 'Files' section and on the right is the 'Resources' section. In the 'Files' section, under 'New,' you can create a blank VI, project, or module-specific project, or you can select the 'More' folder to start from one of the many LabVIEW templates. In the 'Resources' section, you can find additional tools for getting started, ranging from online discussion forums to a comprehensive library of example programs that are shipped with LabVIEW.

The front panel window

When a user opens a new or an existing VI, the front panel window of the VI appears and it now functions as the graphical user interface of the VI. The